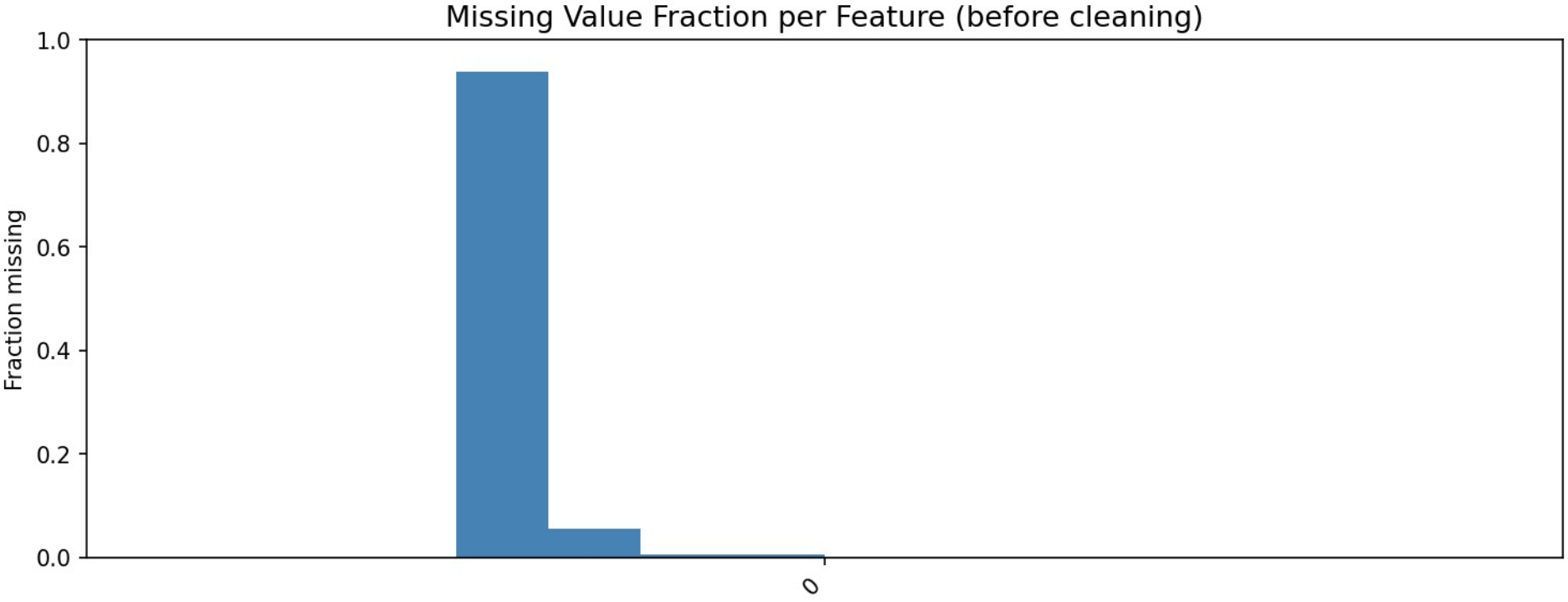
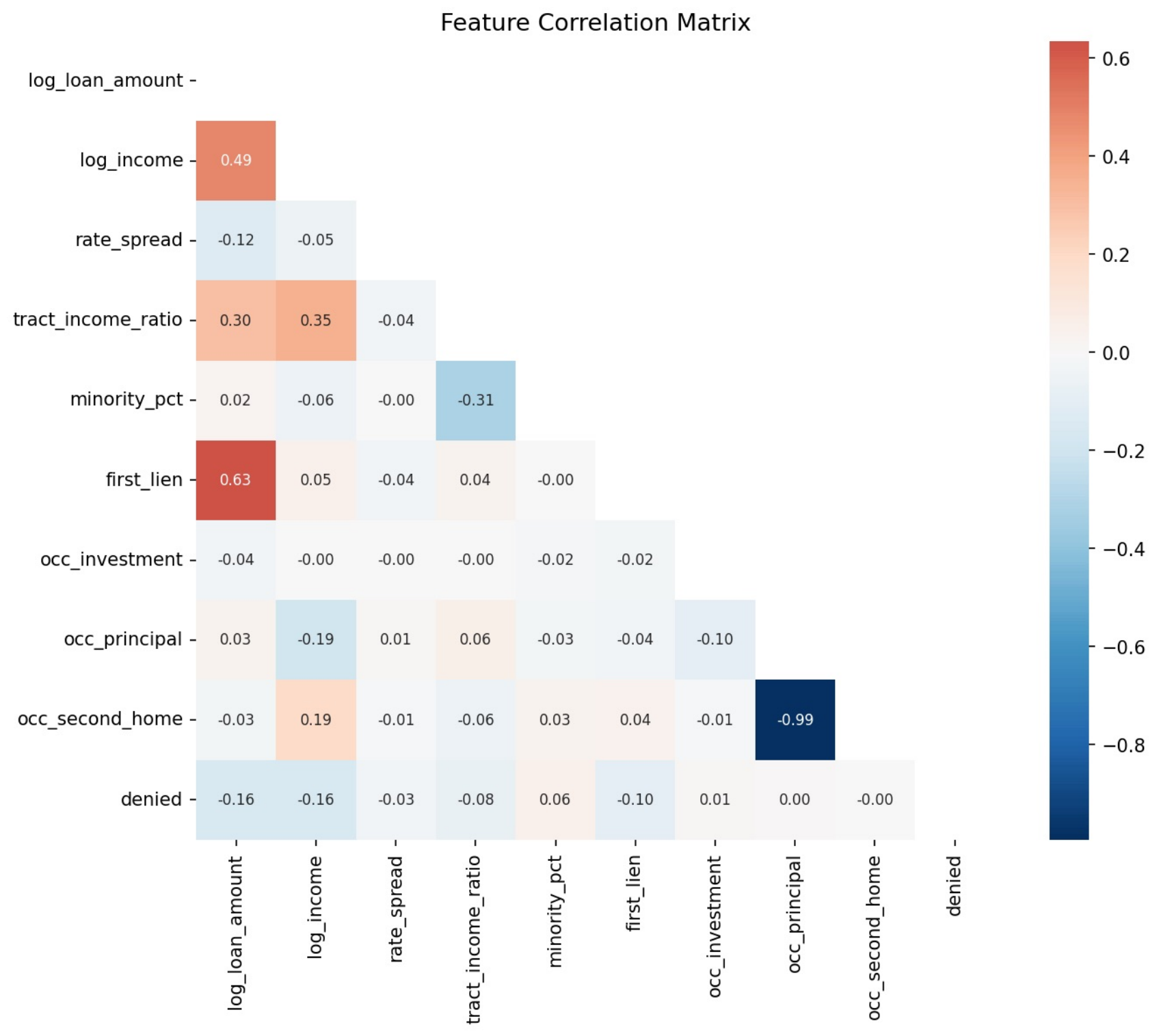
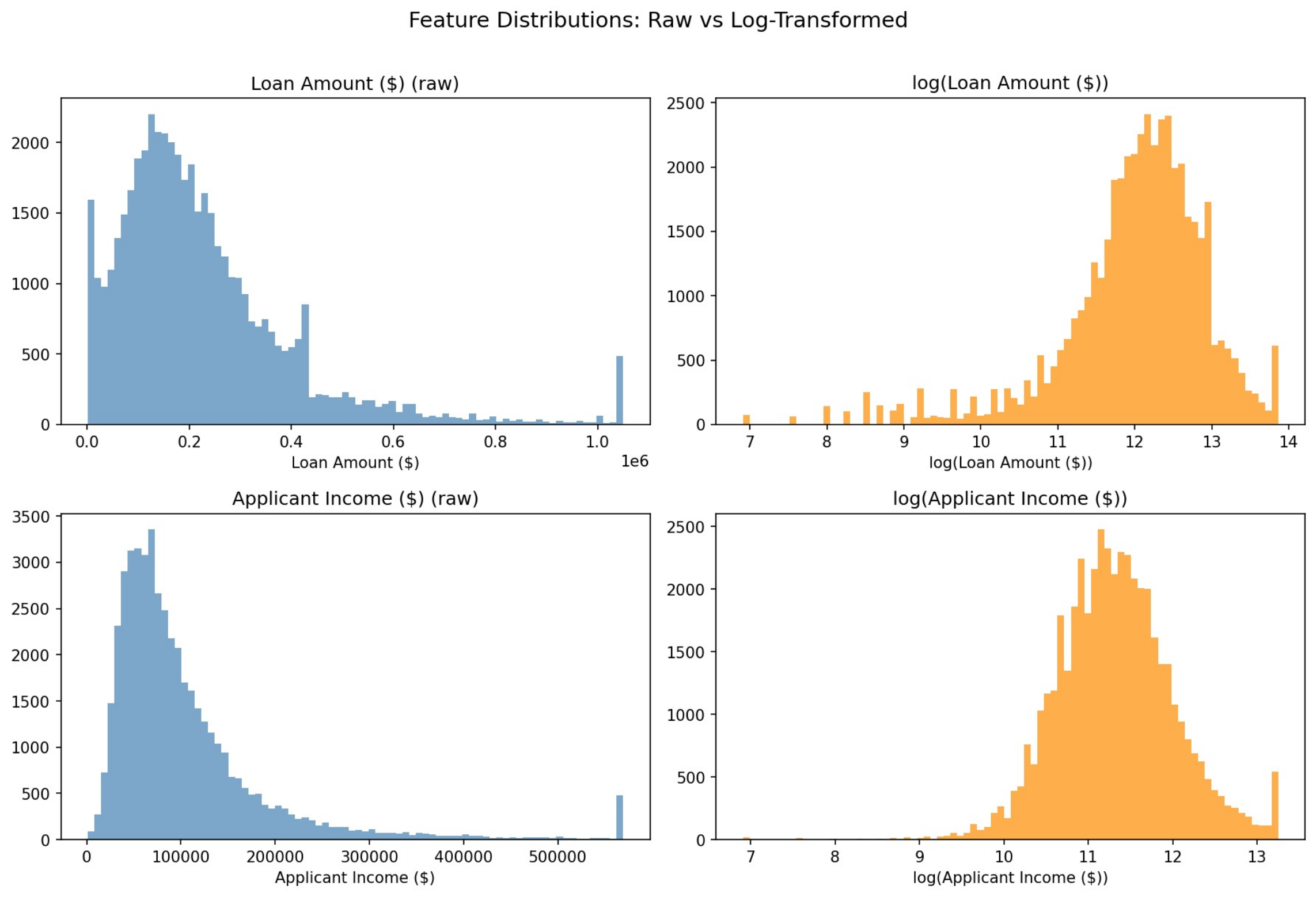
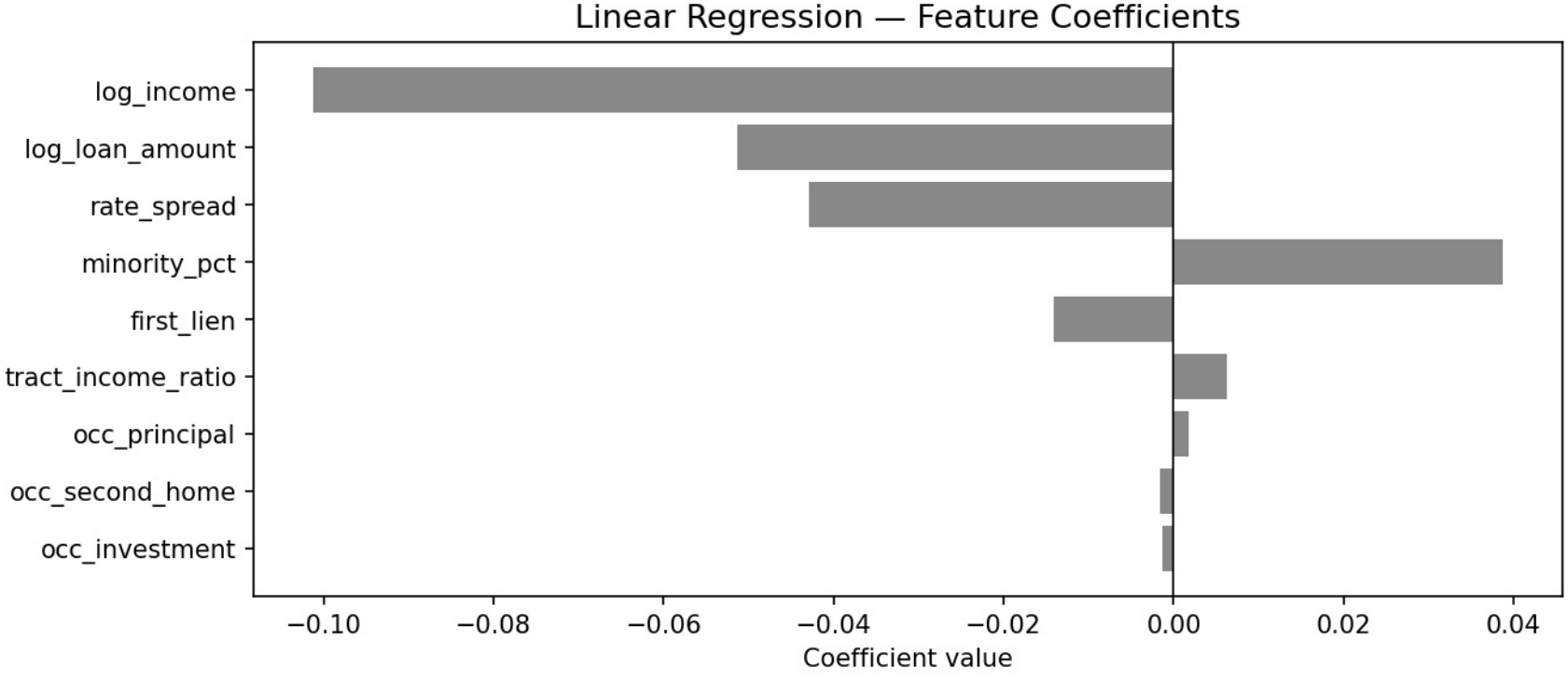
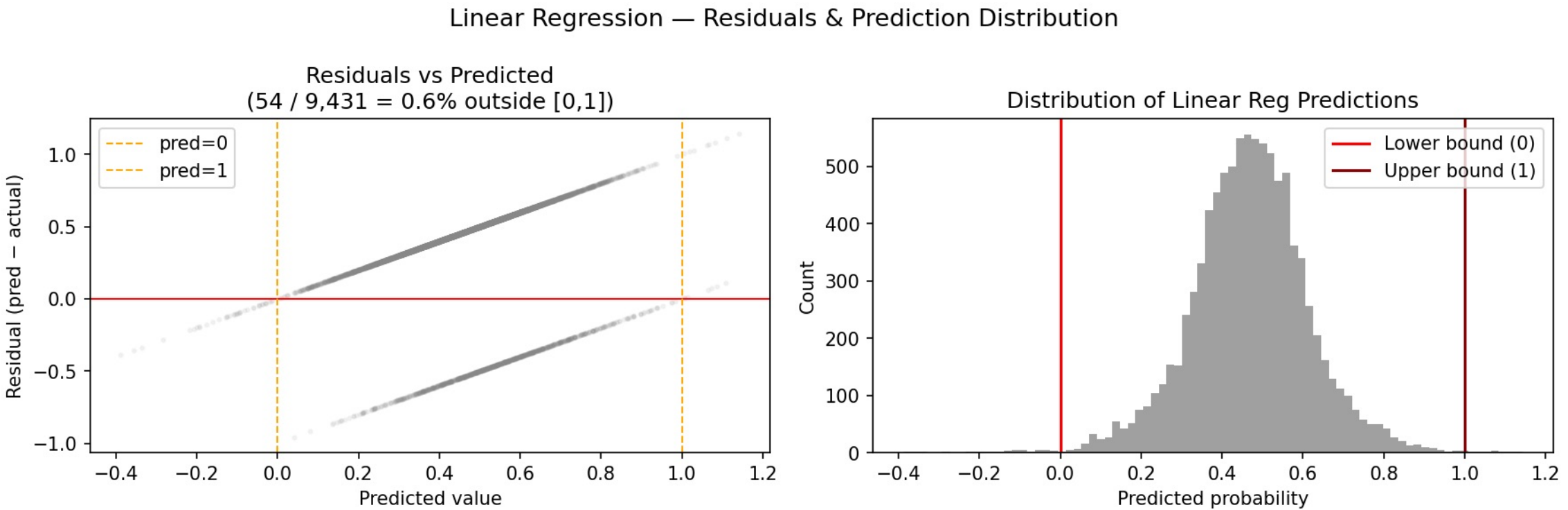

Missing Value Fraction per Feature (before cleaning)
Fraction missing
1.0
0.8
0.6
0.4
0.2
0.0
0

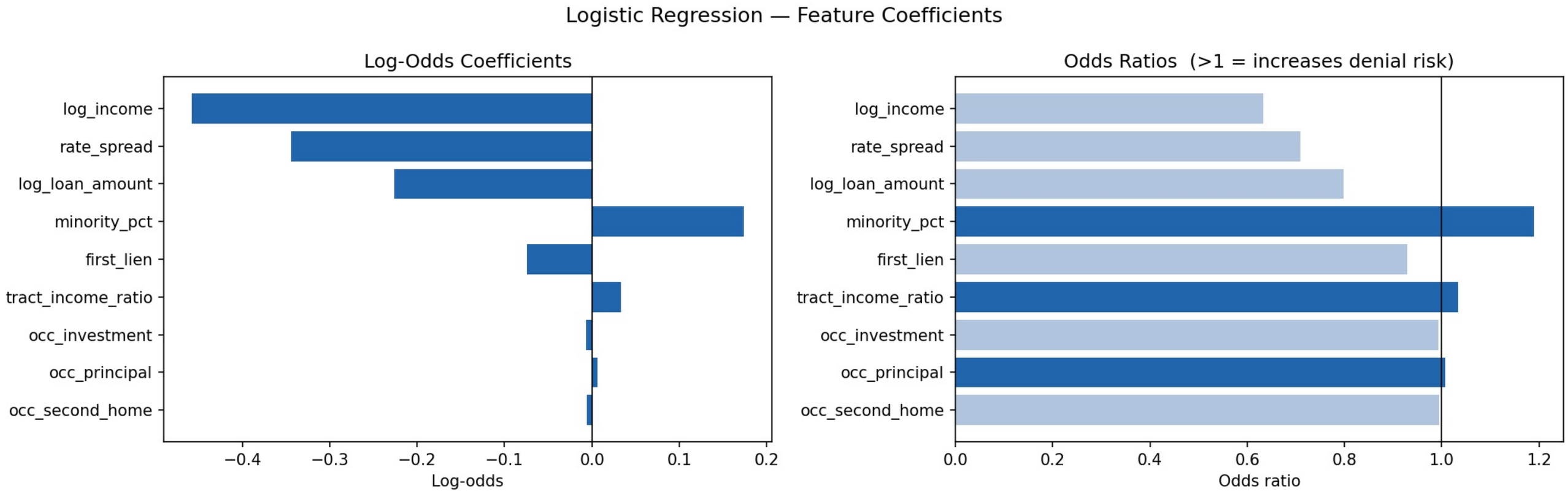

Feature Correlation Matrix
log_loan_amount
log_income
rate_spread
tract_income_ratio
minority_pct
first_lien
occ_investment
occ_principal
occ_second_home
denied
0.49
-0.12 -0.05
0.30 0.35 -0.04
0.02 -0.06 -0.00 -0.31
0.63 0.05 -0.04 0.04 -0.00
-0.04 -0.00 -0.00 -0.00 -0.02 -0.02
0.03 -0.19 0.01 0.06 -0.03 -0.04 -0.10
-0.03 0.19 -0.01 -0.06 0.03 0.04 -0.01 -0.99
-0.16 -0.16 -0.03 -0.08 0.06 -0.10 0.01 0.00 -0.00
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8

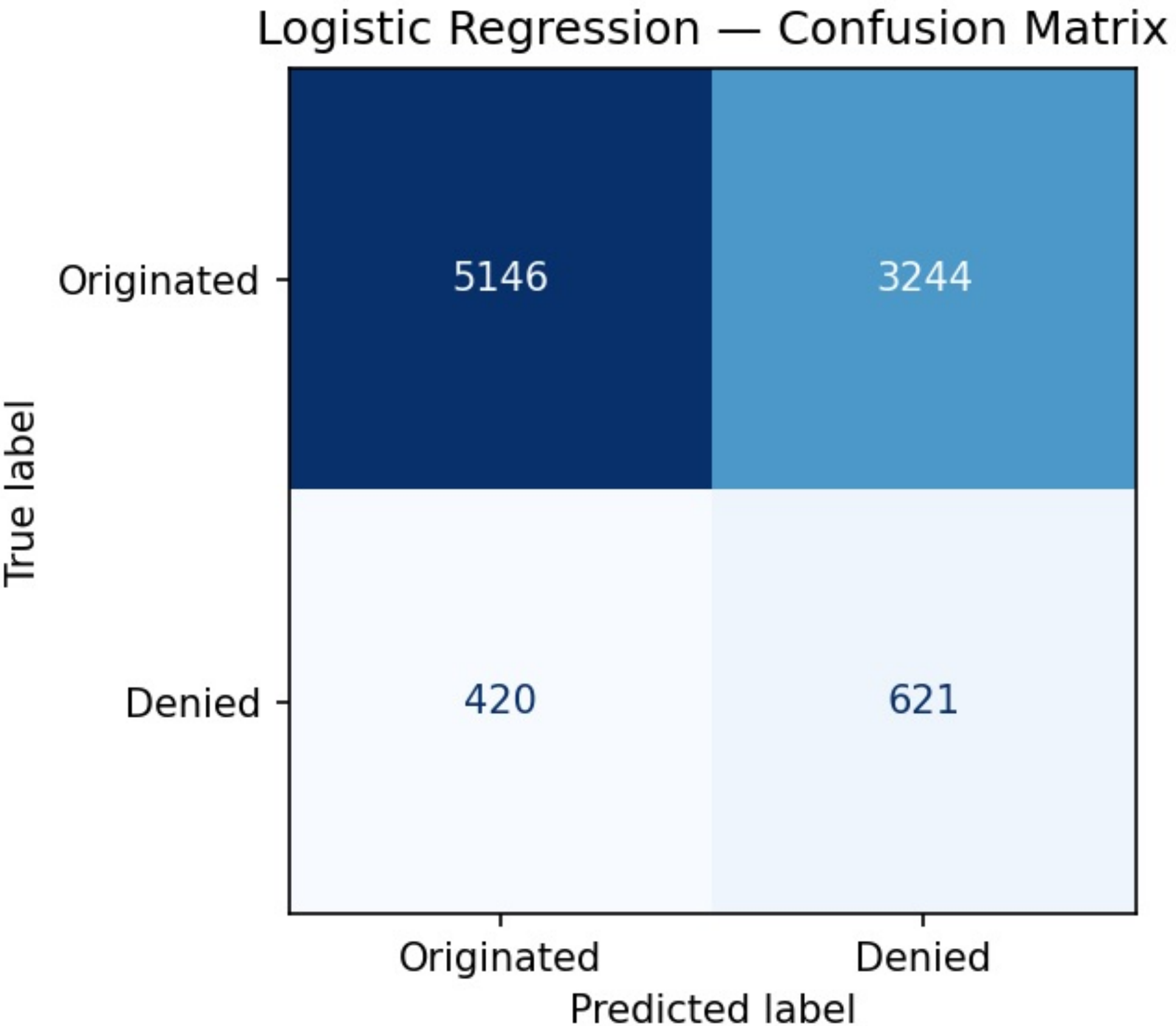

Feature Distributions: Raw vs Log-Transformed
Loan Amount ($) (raw)
Loan Amount ($)
1e6
log(Loan Amount ($))
log(Loan Amount ($))
Applicant Income ($) (raw)
Applicant Income ($)
log(Applicant Income ($))
log(Applicant Income ($))

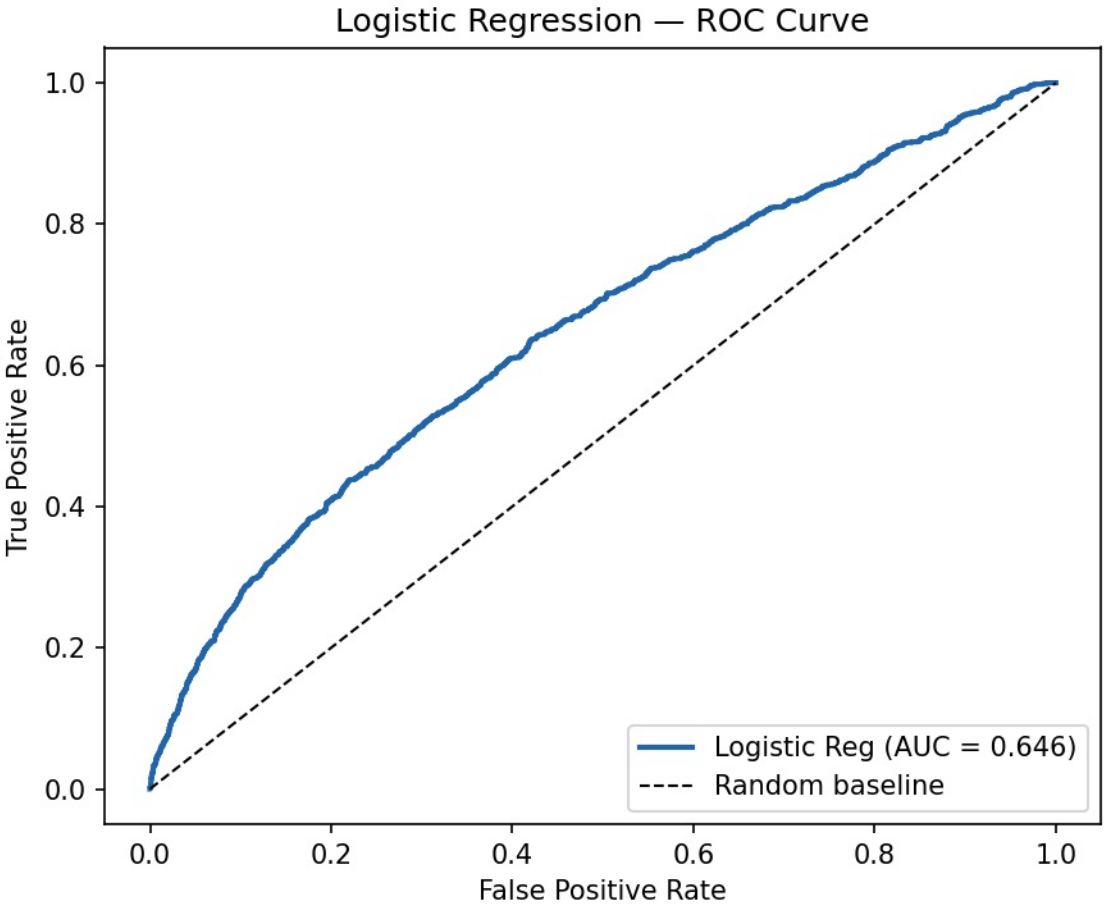

Linear Regression — Feature Coefficients
log_income
log_loan_amount
rate_spread
minority_pct
first_lien
tract_income_ratio
occ_principal
occ_second_home
occ_investment
−0.10
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
Coefficient value

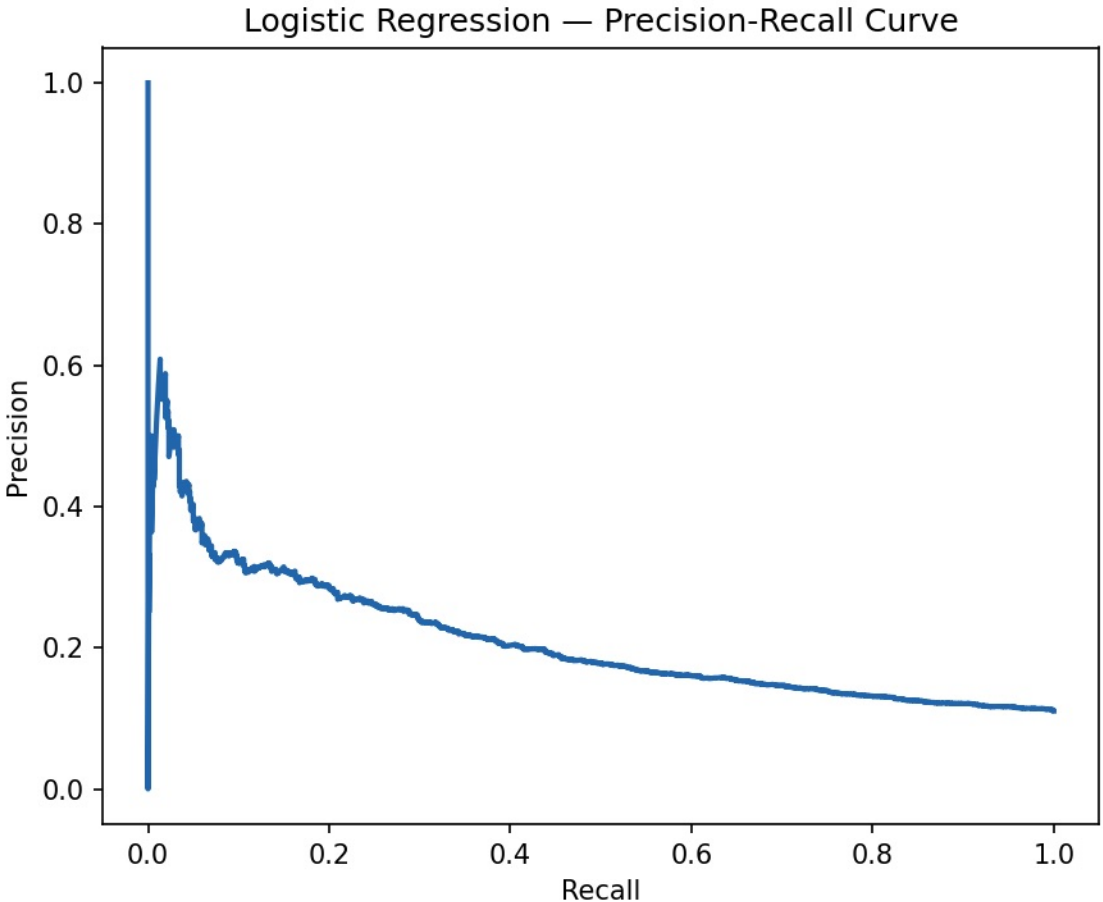
Linear Regression — Confusion Matrix
Originated
Denied
True label
5159
3231
420
621
Originated
Denied
Predicted label

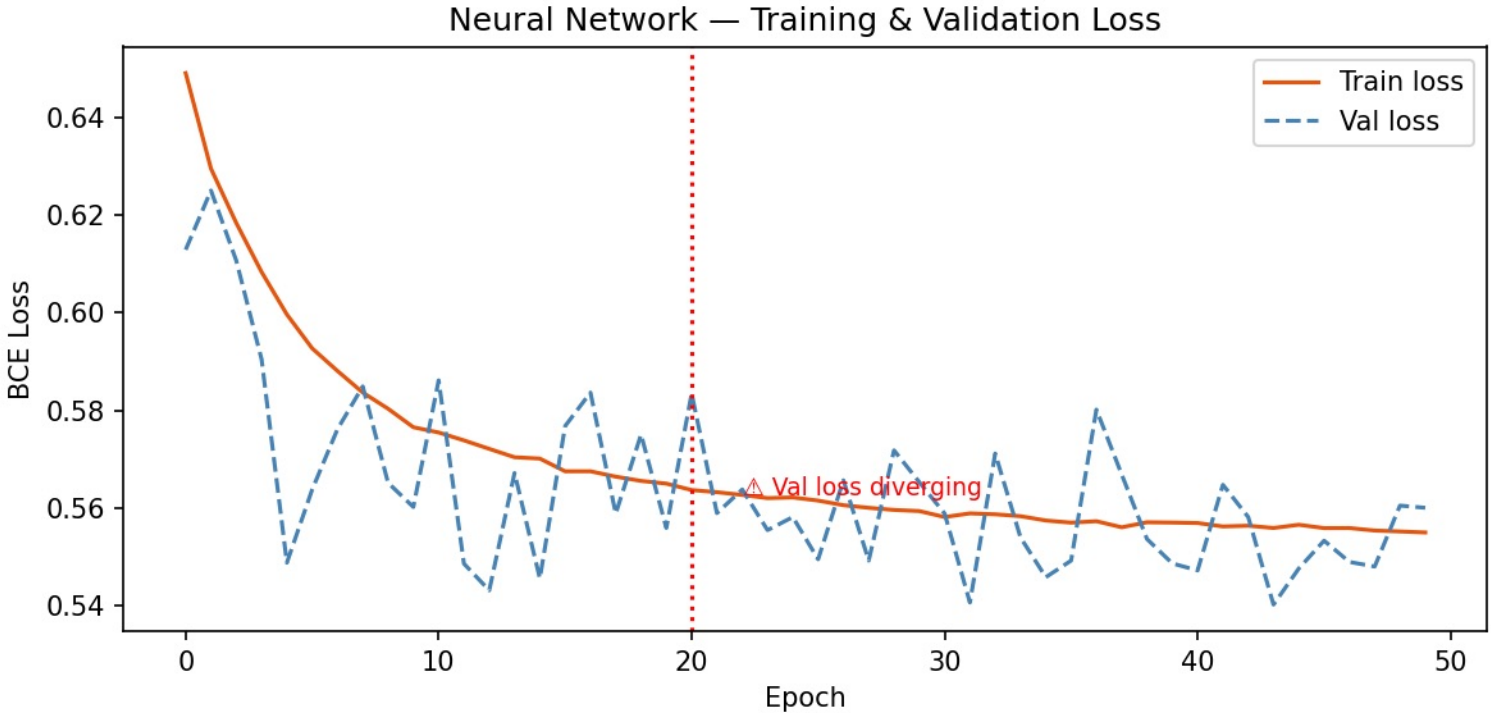

Linear Regression — Residuals & Prediction Distribution
Residuals vs Predicted
(54 / 9,431 = 0.6% outside [0,1])
pred=0
pred=1
Residual (pred − actual)
Predicted value
1.0
0.5
0.0
−0.5
−1.0
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Distribution of Linear Reg Predictions
Lower bound (0)
Upper bound (1)
Count
Predicted probability
500
400
300
200
100
0

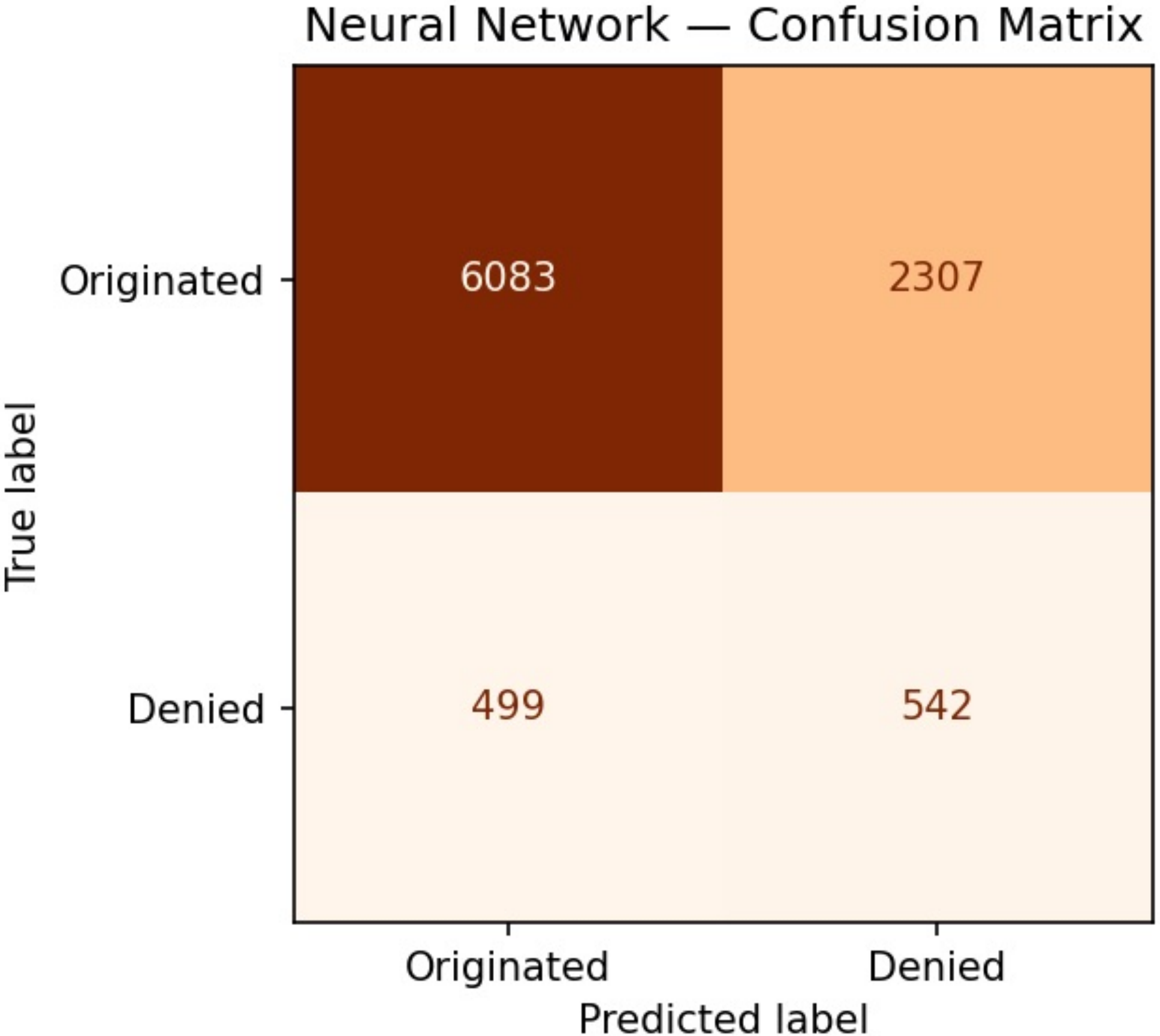

Logistic Regression — Feature Coefficients
Log-Odds Coefficients
log_income
rate_spread
log_loan_amount
minority_pct
first_lien
tract_income_ratio
occ_investment
occ_principal
occ_second_home
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
Log-odds
Odds Ratios (>1 = increases denial risk)
log_income
rate_spread
log_loan_amount
minority_pct
first_lien
tract_income_ratio
occ_investment
occ_principal
occ_second_home
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Odds ratio

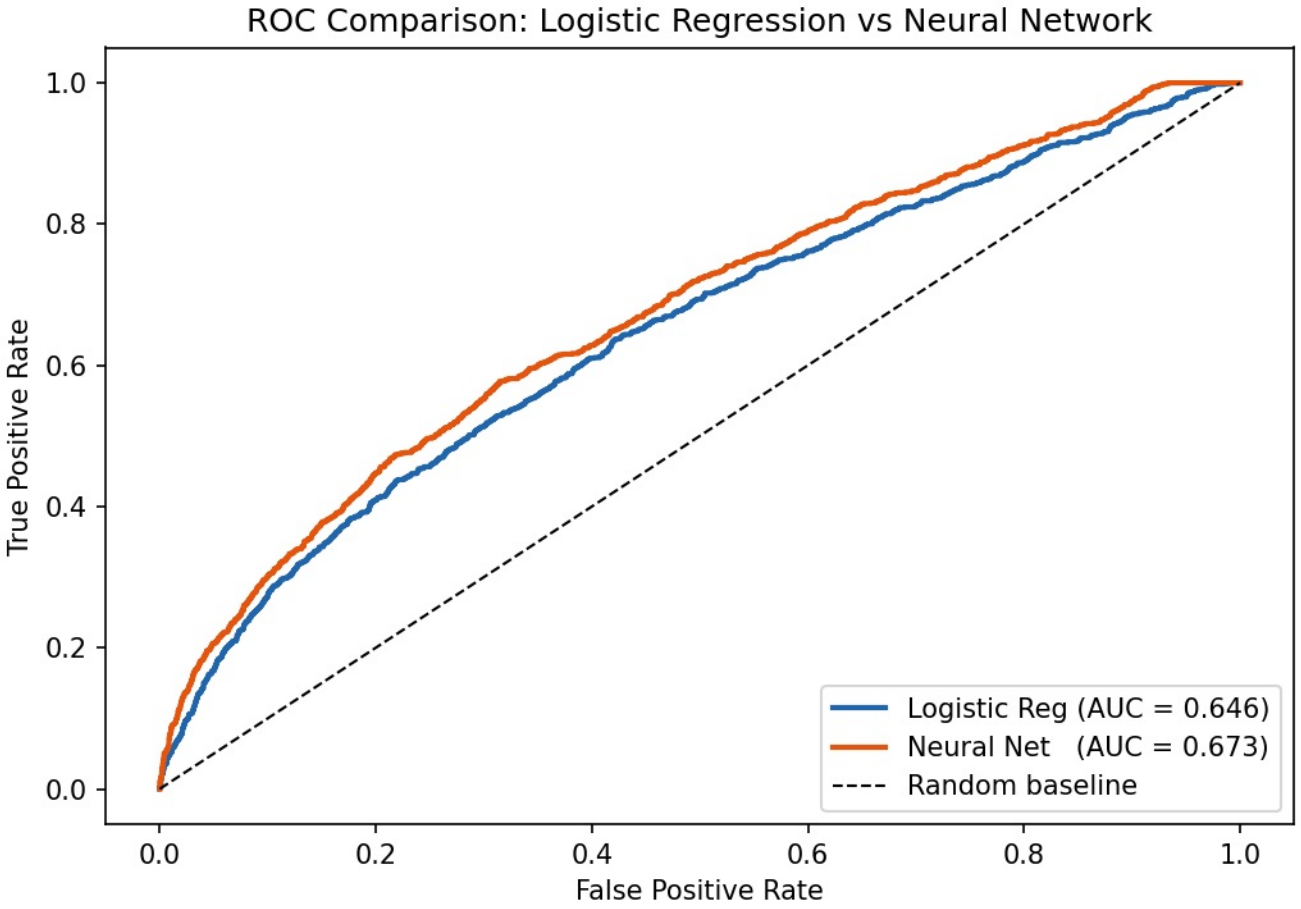

Logistic Regression — Confusion Matrix
Originated
Denied
True label
5146
3244
420
621
Originated
Denied
Predicted label


Logistic Regression — ROC Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Reg (AUC = 0.646)
Random baseline

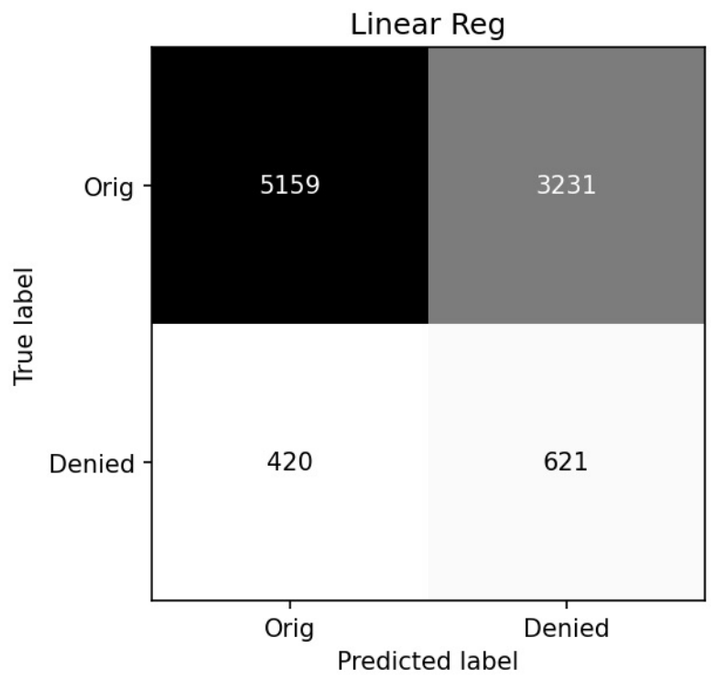

Logistic Regression — Precision-Recall Curve
Precision
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

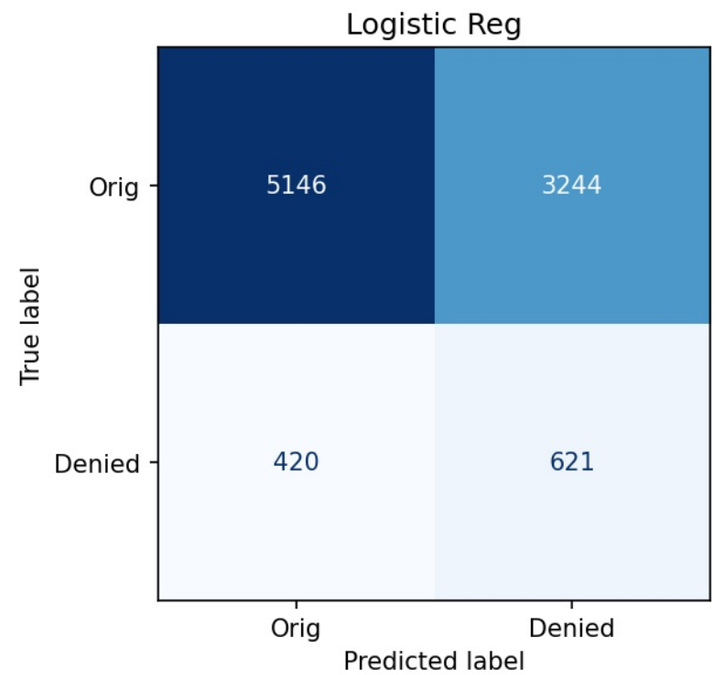

Neural Network — Training & Validation Loss
Train loss
Val loss
⚠ Val loss diverging
BCE Loss
0.64
0.62
0.60
0.58
0.56
0.54
0
10
20
30
40
50
Epoch

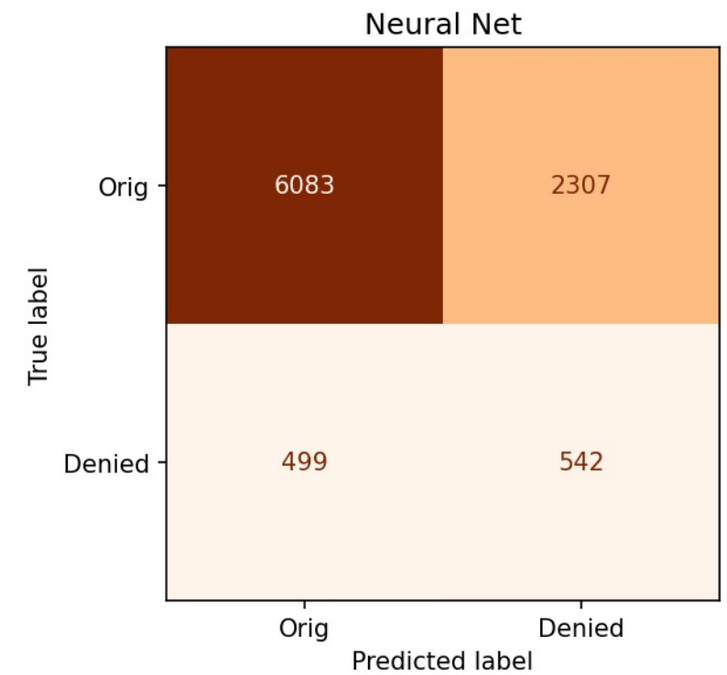

Neural Network — Confusion Matrix
Originated
Denied
True label
6083
2307
499
542
Originated
Denied
Predicted label

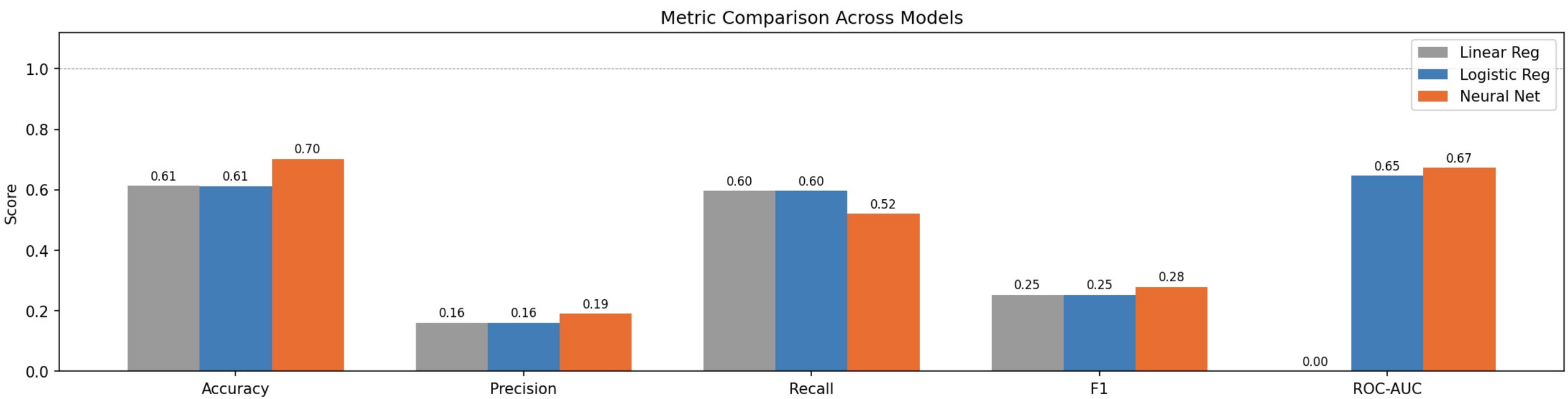

ROC Comparison: Logistic Regression vs Neural Network
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Reg (AUC = 0.646)
Neural Net (AUC = 0.673)
Random baseline

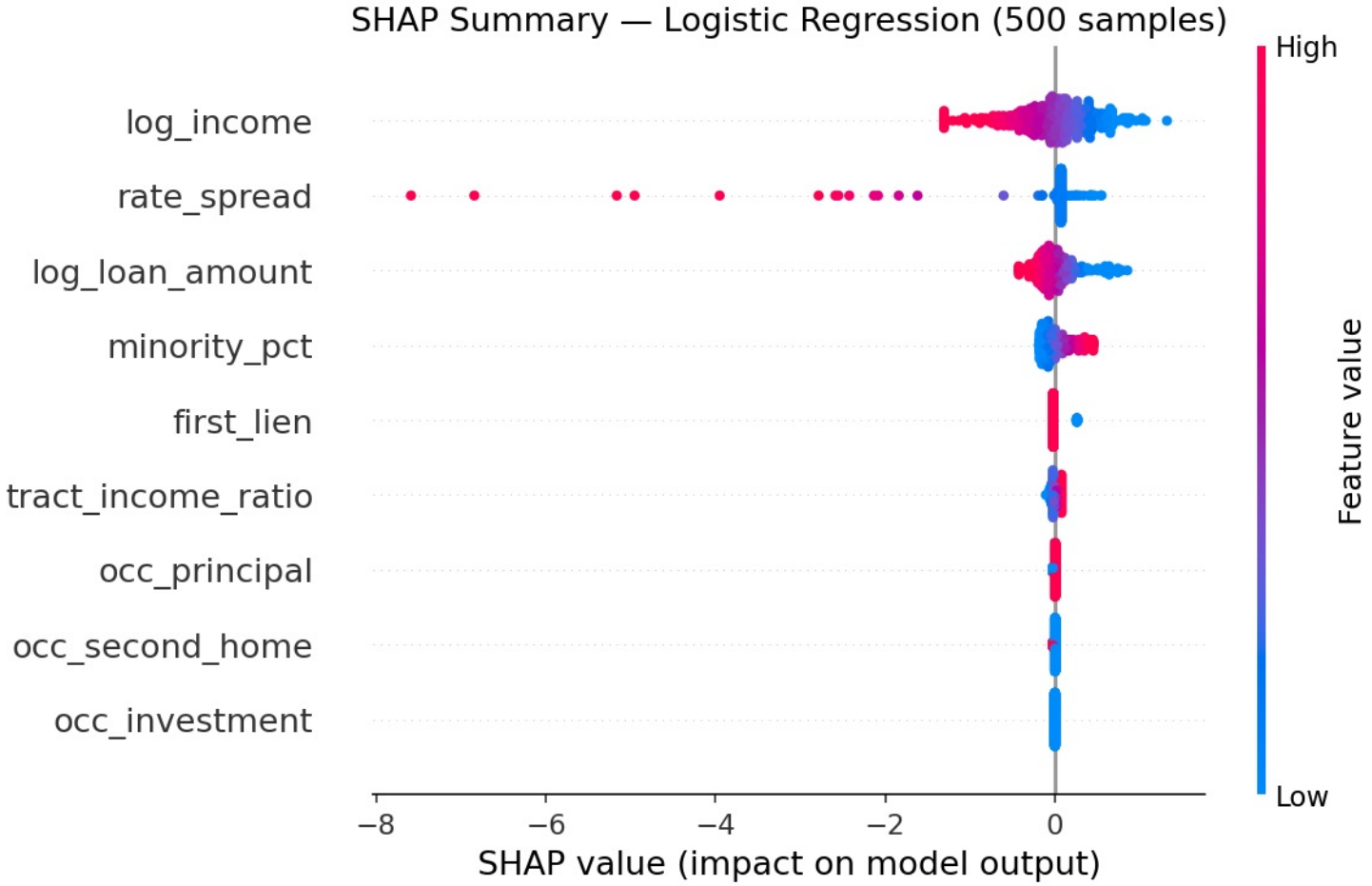
Mortgage Denial Model Comparison Dashboard

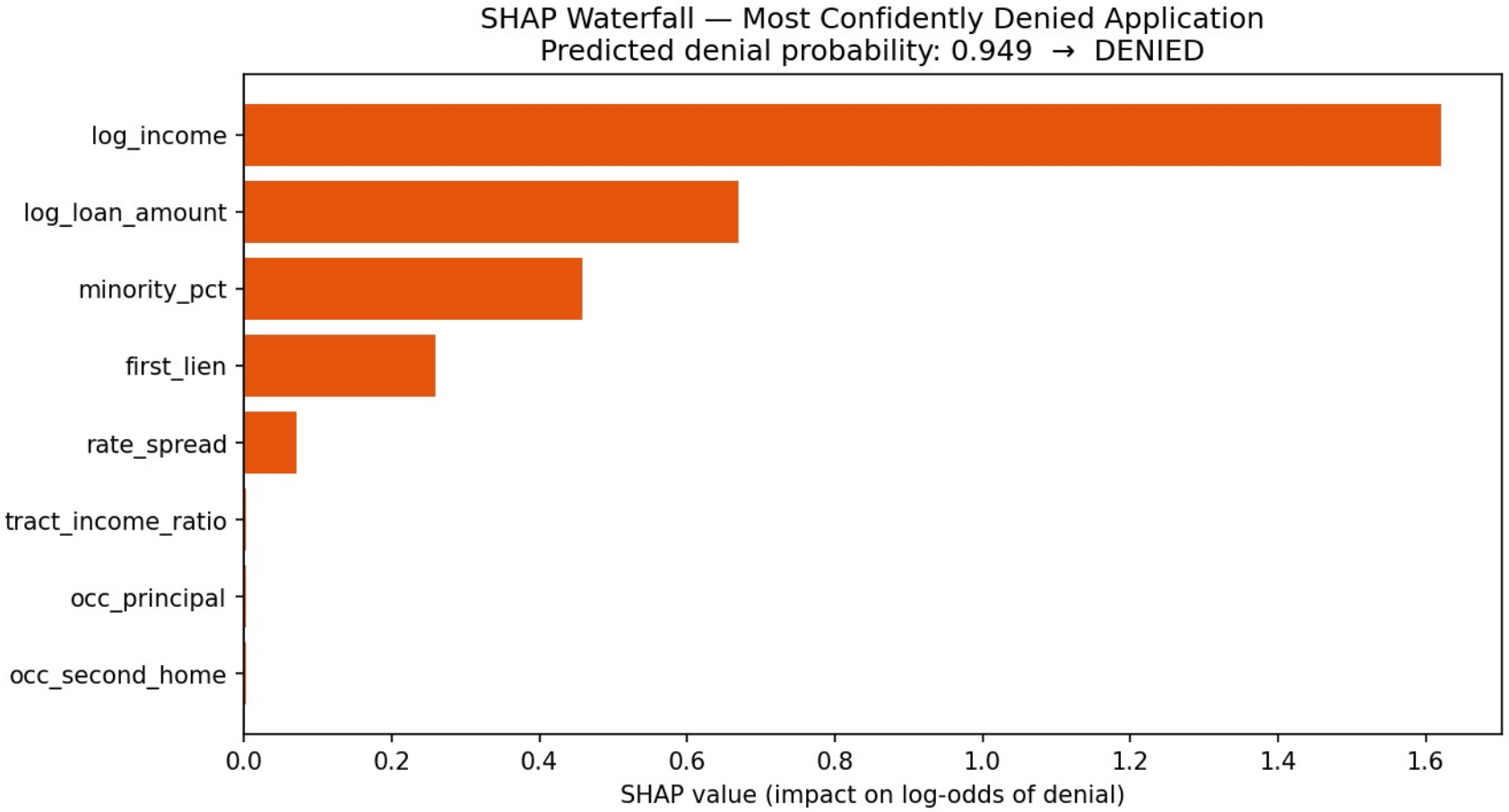









SHAP Summary — Logistic Regression (500 samples)
log_income
rate_spread
log_loan_amount
minority_pct
first_lien
tract_income_ratio
occ_principal
occ_second_home
occ_investment
−8
−6
−4
−2
0
SHAP value (impact on model output)
High
Feature value
Low


SHAP Waterfall — Most Confidently Denied Application
Predicted denial probability: 0.949 → DENIED
log_income
log_loan_amount
minority_pct
first_lien
rate_spread
tract_income_ratio
occ_principal
occ_second_home
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
SHAP value (impact on log-odds of denial)